Powers and Square Roots

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KEY WORDS power *(taupū)* · base *(pūtake)* · square number *(tau pūrua)* · square root *(pūtakerua)*

A Powers of 2 ●○○

Remember: The power tells you how many times the base is multiplied by itself: $2^3 = 2 \times 2 \times 2 = 8$.

1. Work out each power of 2.

a) 2^2 = _____ **b)** 2^4 = _____ **c)** 2^3 = _____ **d)** 2^5 = _____

e) 2^1 = _____ **f)** 2^6 = _____ **g)** 2^8 = _____ **h)** 2^{10} = _____

B Working out powers ●○○

Watch out: 3^2 means $3 \times 3 = 9$. It does not mean 3×2.

2. Work out.

a) 3^2 = _____ **b)** 5^3 = _____ **c)** 10^4 = _____ **d)** 4^3 = _____

e) 7^2 = _____ **f)** 1^9 = _____ **g)** 6^2 = _____ **h)** 3^4 = _____

i) $(-2)^3$ = _____ **j)** 0.5^2 = _____ **k)** 20^2 = _____ **l)** $9^2 - 2^3$ = _____

C The power of zero ●●○

Remember: Any number (except 0) to the power of 0 is 1.

3. Work out.

a) 5^0 = _____ **b)** 12^0 = _____ **c)** $(-3)^0$ = _____ **d)** $2^0 + 3$ = _____

e) 4×7^0 = _____ **f)** $10^0 + 10^1$ = _____ **g)** $6^0 \times 6^1$ = _____ **h)** $(2 + 5)^0$ = _____

D Negative powers ●●●

Remember: A negative power means "one over": $2^{-3} = \frac{1}{2^3} = \frac{1}{8}$.

4. Write each power as a fraction.

a) 2^{-1} **b)** 3^{-2} **c)** 5^{-2}

d) 2^{-4} **e)** 10^{-3} **f)** 4^{-2}

g) 7^{-1} **h)** 3^{-3} **i)** 6^{-2}

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5. Write each fraction as a power with a negative exponent.

a) $\frac{1}{25}$ as a power of 5 **b)** $\frac{1}{8}$ as a power of 2 **c)** $\frac{1}{100}$ as a power of 10

d) $\frac{1}{27}$ as a power of 3 **e)** $\frac{1}{49}$ as a power of 7 **f)** $\frac{1}{64}$ as a power of 2

E Square roots ●●○

Remember: The square root undoes squaring: $\sqrt{49} = 7$ because $7^2 = 49$.

6. Work out.

a) $\sqrt{49}$ **b)** $\sqrt{121}$ **c)** $\sqrt{400}$ **d)** $\sqrt{0.25}$

e) $\sqrt{16} + \sqrt{9}$ **f)** $\sqrt{16 + 9}$ **g)** $\sqrt{10^2 - 6^2}$ **h)** $(\sqrt{11})^2$

7. Without a calculator, say which two whole numbers each square root lies between, and which of them it is closer to.

a) $\sqrt{30}$ **b)** $\sqrt{50}$ **c)** $\sqrt{12}$

d) $\sqrt{90}$ **e)** $\sqrt{70}$ **f)** $\sqrt{150}$

Extension

8. Find n if $2^n = 64$.

9. Find n if $3^n = \frac{1}{9}$.

10. Which is larger, 2^5 or 5^2? Show your working.

11. Find n if $2^n \times 2^3 = 2^7$. Think about what happens to the exponents when powers of the same base are multiplied.

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Powers and Square Roots

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KEY WORDS power *(taupū)* · base *(pūtake)* · square number *(tau pūrua)* · square root *(pūtakerua)*

A Powers of 2 ●○○

Remember: The power tells you how many times the base is multiplied by itself: $2^3 = 2 \times 2 \times 2 = 8$.

1. Work out each power of 2.

a) 2^3 = _____ **b)** 2^2 = _____ **c)** 2^5 = _____ **d)** 2^4 = _____

e) 2^1 = _____ **f)** 2^7 = _____ **g)** 2^9 = _____ **h)** 2^{10} = _____

B Working out powers ●○○

Watch out: 3^2 means $3 \times 3 = 9$. It does not mean 3×2.

2. Work out.

a) 4^2 = _____ **b)** 2^5 = _____ **c)** 10^3 = _____ **d)** 5^3 = _____

e) 8^2 = _____ **f)** 1^7 = _____ **g)** 9^2 = _____ **h)** 3^3 = _____

i) $(-3)^3$ = _____ **j)** 0.2^2 = _____ **k)** 30^2 = _____ **l)** $7^2 - 3^3$ = _____

C The power of zero ●●○

Remember: Any number (except 0) to the power of 0 is 1.

3. Work out.

a) 7^0 = _____ **b)** 25^0 = _____ **c)** $(-8)^0$ = _____ **d)** $3^0 + 5$ = _____

e) 6×4^0 = _____ **f)** $10^0 + 10^2$ = _____ **g)** $9^0 \times 9^1$ = _____ **h)** $(3 + 4)^0$ = _____

D Negative powers ●●●

Remember: A negative power means "one over": $2^{-3} = \frac{1}{2^3} = \frac{1}{8}$.

4. Write each power as a fraction.

a) 3^{-1} **b)** 2^{-2} **c)** 4^{-2}

d) 2^{-5} **e)** 10^{-2} **f)** 5^{-3}

g) 9^{-1} **h)** 3^{-4} **i)** 8^{-2}

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5. Write each fraction as a power with a negative exponent.

a) $\frac{1}{16}$ as a power of 4 **b)** $\frac{1}{16}$ as a power of 2 **c)** $\frac{1}{10000}$ as a power of 10

d) $\frac{1}{9}$ as a power of 3 **e)** $\frac{1}{36}$ as a power of 6 **f)** $\frac{1}{128}$ as a power of 2

E Square roots ●●○

Remember: The square root undoes squaring: $\sqrt{49} = 7$ because $7^2 = 49$.

6. Work out.

a) $\sqrt{64}$ **b)** $\sqrt{144}$ **c)** $\sqrt{900}$ **d)** $\sqrt{0.09}$

e) $\sqrt{25} + \sqrt{144}$ **f)** $\sqrt{25 + 144}$ **g)** $\sqrt{5^2 - 3^2}$ **h)** $(\sqrt{13})^2$

7. Without a calculator, say which two whole numbers each square root lies between, and which of them it is closer to.

a) $\sqrt{40}$ **b)** $\sqrt{20}$ **c)** $\sqrt{60}$

d) $\sqrt{110}$ **e)** $\sqrt{75}$ **f)** $\sqrt{130}$

★ Extension

8. Find n if $3^n = 81$.

9. Find n if $2^n = \frac{1}{32}$.

10. Which is larger, 3^4 or 4^3? Show your working.

11. Find n if $3^n \times 3^2 = 3^7$. Think about what happens to the exponents when powers of the same base are multiplied.

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